

Hematology and Oncology

“Of all that is written, I love only what a person has written with his own blood.”

—Friedrich Nietzsche

“I used to get stressed out, but my cancer has put everything into perspective.”

—Delta Goodrem

“The best blood will at some time get into a fool or a mosquito.”

—Austin O’Malley

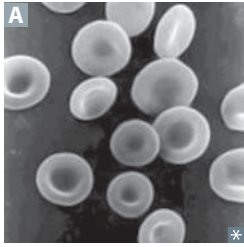
“Carcinoma works cunningly from the inside out. Detection and treatment often work more slowly and gropingly, from the outside in.”

—Christopher Hitchens

Study tip: When reviewing oncologic drugs, focus on mechanisms and side effects rather than details of clinical uses, which may be lower yield.

▶ Anatomy	386
▶ Physiology	389
▶ Pathology	394
▶ Pharmacology	413

▶ HEMATOLOGY AND ONCOLOGY—ANATOMY

Erythrocyte

Carries O_2 to tissues and CO_2 to lungs.

Anucleate and lacks organelles; biconcave **A**, with large surface area-to-volume ratio for rapid gas exchange. Life span of 120 days. Source of energy is glucose (90% used in glycolysis, 10% used in HMP shunt). Membrane contains Cl^-/HCO_3^- antiporter, which allows RBCs to export HCO_3^- and transport CO_2 from the periphery to the lungs for elimination.

Eryth = red; *cyte* = cell.

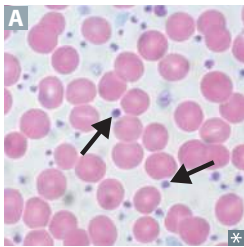
Erythrocytosis = polycythemia = ↑ hematocrit.

Anisocytosis = varying sizes.

Poikilocytosis = varying shapes.

Reticulocyte = immature RBC; reflects erythroid proliferation.

Bluish color on Wright-Giemsa stain of reticulocytes represents residual ribosomal RNA.

Thrombocyte (platelet)

Involved in 1° hemostasis. Small cytoplasmic fragment **A** derived from megakaryocytes. Life span of 8–10 days. When activated by endothelial injury, aggregates with other platelets and interacts with fibrinogen to form platelet plug. Contains dense granules (ADP, Ca^{2+}) and α granules (vWF, fibrinogen, fibronectin). Approximately $\frac{1}{3}$ of platelet pool is stored in the spleen.

Thrombocytopenia or ↓ platelet function results in petechiae.

vWF receptor: GpIb.

Fibrinogen receptor: GpIIb/IIIa.

Leukocyte

Divided into granulocytes (neutrophil, eosinophil, basophil, mast cell) and mononuclear cells (monocytes, lymphocytes). WBC differential count from highest to lowest (normal ranges per USMLE):

Neutrophils (~ 60%)

Lymphocytes (~ 30%)

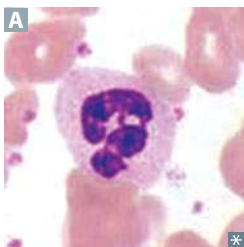
Monocytes (~ 6%)

Eosinophils (~ 3%)

Basophils (~ 1%)

Leuk = white; *cyte* = cell.

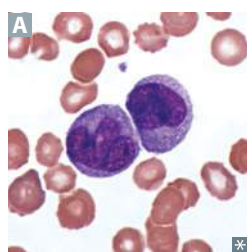
Neutrophils **L**ike **M**aking **E**verything **B**etter.

Neutrophil

Acute inflammatory response cell. Increased in bacterial infections. Phagocytic. Multilobed nucleus **A**. Specific granules contain leukocyte alkaline phosphatase (LAP), collagenase, lysozyme, and lactoferrin. Azurophilic granules (lysosomes) contain proteinases, acid phosphatase, myeloperoxidase, and β -glucuronidase.

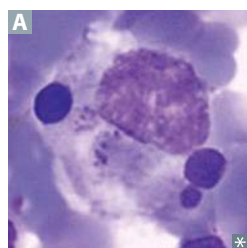
Hypersegmented neutrophils (nucleus has 6+ lobes) are seen in vitamin B_{12} / folate deficiency. ↑ band cells (immature neutrophils) reflect states of ↑ myeloid proliferation (bacterial infections, CML).

Important neutrophil chemotactic agents: C5a, IL-8, LTB_4 , kallikrein, platelet-activating factor.

Monocyte

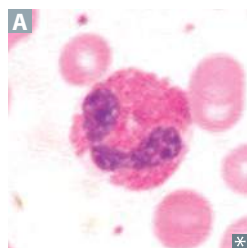
Differentiates into macrophage in tissues.
Large, kidney-shaped nucleus **A**. Extensive
“frosted glass” cytoplasm.

Mono = one (nucleus); *cyte* = cell.
Found in blood.

Macrophage

Phagocytoses bacteria, cellular debris, and
senescent RBCs. Long life in tissues.
Macrophages differentiate from circulating
blood monocytes **A**. Activated by γ -interferon.
Can function as antigen-presenting cell via
MHC II.

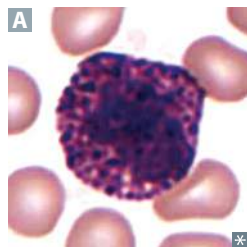
Macro = large; *phage* = eater.
Found in tissue. Name differs in each tissue type
(eg, Kupffer cells in the liver, histiocytes in
connective tissue).
Important component of granuloma formation
(eg, TB, sarcoidosis).
Lipid A from bacterial LPS binds CD14 on
macrophages to initiate septic shock.

Eosinophil

Defends against helminthic infections (major
basic protein). Bilobate nucleus. Packed
with large eosinophilic granules of uniform
size **A**. Highly phagocytic for antigen-
antibody complexes.

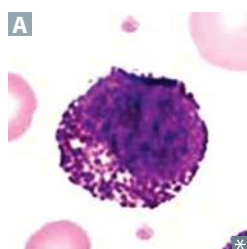
Produces histaminase, major basic protein
(MBP, a helminthotoxin), eosinophil
peroxidase, eosinophil cationic protein, and
eosinophil-derived neurotoxin.

Eosin = pink dye; *philic* = loving.
Causes of eosinophilia = **NAACP**:
Neoplasia
Asthma
Allergic processes
Chronic adrenal insufficiency
Parasites (invasive)

Basophil

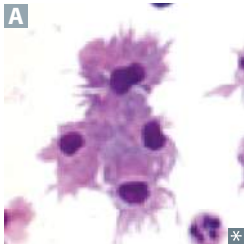
Mediates allergic reaction. Densely basophilic
granules **A** contain heparin (anticoagulant)
and histamine (vasodilator). Leukotrienes
synthesized and released on demand.

Basophilic—staining readily with **basic** stains.
Basophilia is uncommon, but can be a sign of
myeloproliferative disease, particularly CML.

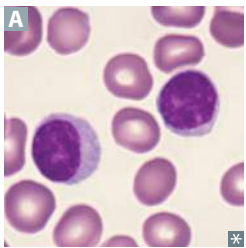
Mast cell

Mediates allergic reaction in local tissues.
Mast cells contain basophilic granules **A** and
originate from the same precursor as basophils
but are not the same cell type. Can bind the
Fc portion of IgE to membrane. IgE cross-
links upon antigen binding → degranulation
→ release of histamine, heparin, tryptase, and
eosinophil chemotactic factors.

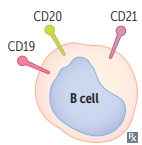
Involved in type I hypersensitivity reactions.
Cromolyn sodium prevents mast cell
degranulation (used for asthma prophylaxis).

Dendritic cell

Highly phagocytic antigen-presenting cell (APC) **A**. Functions as link between innate and adaptive immune systems. Expresses MHC class II and Fc receptors on surface. Called Langerhans cell in the skin.

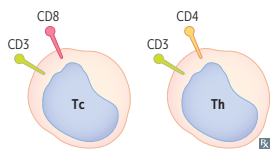
Lymphocyte

Refers to B cells, T cells, and NK cells. B cells and T cells mediate adaptive immunity. NK cells are part of the innate immune response. Round, densely staining nucleus with small amount of pale cytoplasm **A**.

B cell

Part of humoral immune response. Originates from stem cells in bone marrow and matures in marrow. Migrates to peripheral lymphoid tissue (follicles of lymph nodes, white pulp of spleen, unencapsulated lymphoid tissue). When antigen is encountered, B cells differentiate into plasma cells (which produce antibodies) and memory cells. Can function as an APC via MHC II.

B = Bone marrow.

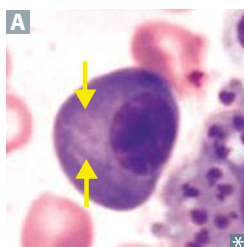
T cell

Mediates cellular immune response. Originates from stem cells in the bone marrow, but matures in the thymus. T cells differentiate into cytotoxic T cells (express CD8, recognize MHC I), helper T cells (express CD4, recognize MHC II), and regulatory T cells. CD28 (costimulatory signal) necessary for T-cell activation. The majority of circulating lymphocytes are T cells (80%).

T is for **T**hymus.

CD4+ helper T cells are the primary target of HIV.

Rule of 8: MHC **II** × CD**4** = **8**;
MHC **I** × CD**8** = **8**.

Plasma cell

Produces large amounts of antibody specific to a particular antigen. “Clock-face” chromatin distribution and eccentric nucleus, abundant RER, and well-developed Golgi apparatus (arrows in **A**). Found in bone marrow and normally do not circulate in peripheral blood.

Multiple myeloma is a plasma cell cancer.

▶ HEMATOLOGY AND ONCOLOGY—PHYSIOLOGY

Fetal erythropoiesis

Fetal erythropoiesis occurs in:

- **Y**olk sac (3–8 weeks)
- **L**iver (6 weeks–birth)
- **S**pleen (10–28 weeks)
- **B**one marrow (18 weeks to adult)

Young **L**iver **S**ynthesizes **B**lood.

Hemoglobin development

Embryonic globins: ζ and ϵ .

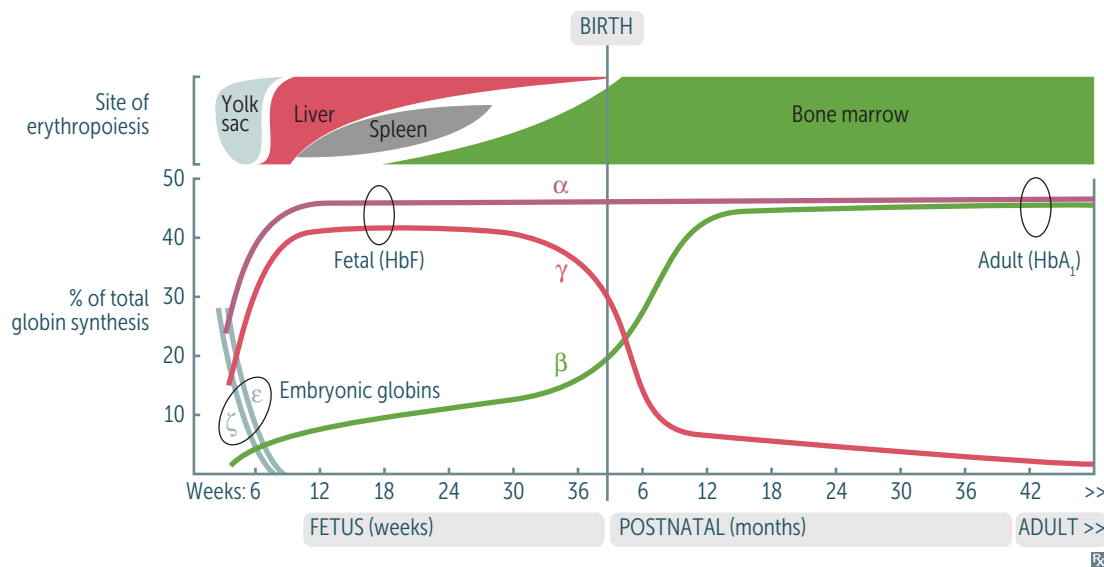
Fetal hemoglobin (HbF) = $\alpha_2\gamma_2$.

Adult hemoglobin (HbA_1) = $\alpha_2\beta_2$.

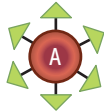
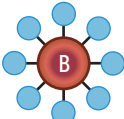
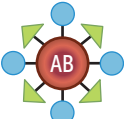
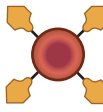
HbF has higher affinity for O_2 due to less avid binding of 2,3-BPG, allowing HbF to extract O_2 from maternal hemoglobin (HbA_1 and HbA_2) across the placenta. HbA_2 ($\alpha_2\delta_2$) is a form of adult hemoglobin present in small amounts.

From fetal to adult hemoglobin:

Alpha **A**lways; **G**amma **G**oes, **B**ecomes **B**eta.



Blood groups

	ABO classification				Rh classification	
	A	B	AB	O	Rh ⁺	Rh ⁻
RBC type						
Group antigens on RBC surface	A 	B 	A & B 	NONE	Rh (D) 	NONE
Antibodies in plasma	Anti-B  IgM	Anti-A  IgM	NONE	Anti-A Anti-B  IgM	NONE	Anti-D  IgG
Clinical relevance	Receive B or AB → hemolytic reaction	Receive A or AB → hemolytic reaction	Universal recipient of RBCs; universal donor of plasma	Receive any non-O → hemolytic reaction Universal donor of RBCs; universal recipient of plasma	Universal recipient of RBCs	Treat mother with anti-D Ig (RhoGAM) during and after each pregnancy to prevent anti-D IgG formation

Rh hemolytic disease of the newborn

IgM does not cross placenta; IgG does cross placenta.

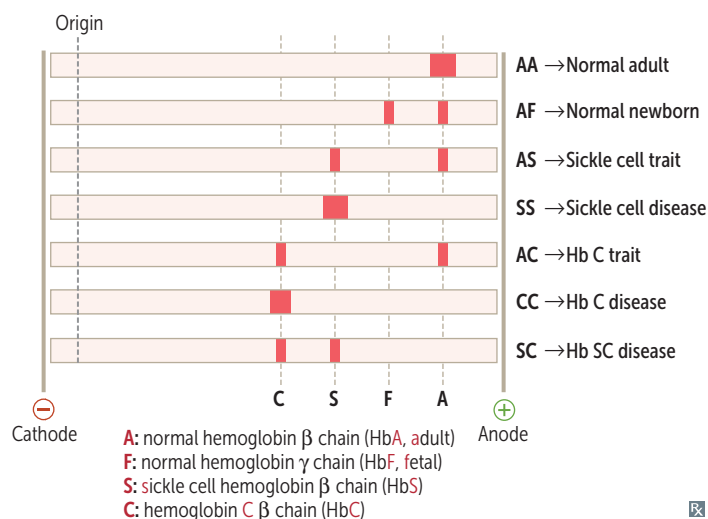
Rh⁻ mothers exposed to fetal Rh⁺ blood (often during delivery) may make anti-D IgG. In subsequent pregnancies, anti-D IgG crosses the placenta → hemolytic disease of the newborn (erythroblastosis fetalis) in the next fetus that is Rh⁺. Administration of anti-D IgG (RhoGAM) to Rh⁻ pregnant women during third trimester and early postpartum period prevents maternal anti-D IgG production.

Rh⁻ mothers have anti-D IgG only if previously exposed to Rh⁺ blood.

ABO hemolytic disease of the newborn

Usually occurs in a type O mother with a type A or B fetus. Can occur in a first pregnancy as maternal anti-A and/or anti-B IgG antibodies may be formed prior to pregnancy. Does not worsen with future pregnancies. Presents as mild jaundice in the neonate within 24 hours of birth; treatment is phototherapy or exchange transfusion.

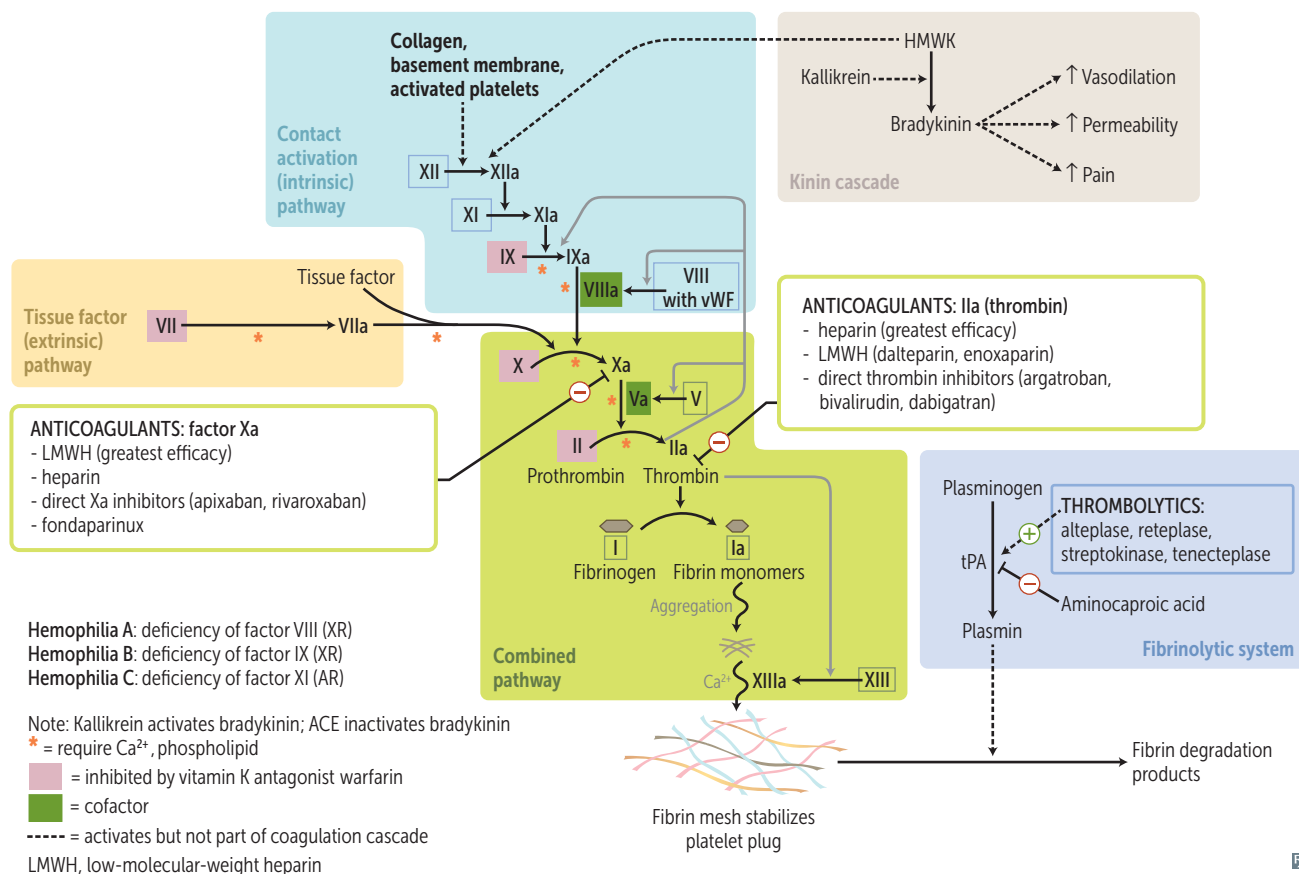
Hemoglobin electrophoresis



On a gel, hemoglobin migrates from the negatively charged cathode to the positively charged anode. HbA migrates the farthest, followed by HbF, HbS, and HbC. This is because the missense mutations in HbS and HbC replace glutamic acid $\ominus$ with valine (neutral) and lysine $\oplus$, respectively, impacting the net protein charge.

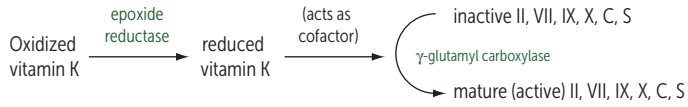
A Fat Santa Claus

Coagulation and kinin pathways



Coagulation cascade components

Procoagulation



Warfarin inhibits the enzyme vitamin K epoxide reductase.

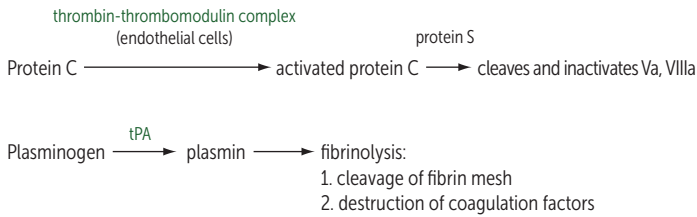
Neonates lack enteric bacteria, which produce vitamin K.

Vitamin K deficiency: ↓ synthesis of factors II, VII, IX, X, protein C, protein S.

vWF carries/protects **VIII** (volksWagen

Factories make **gr8** (great) cars.

Anticoagulation



Antithrombin inhibits activated forms of factors II, VII, IX, X, XI, XII.

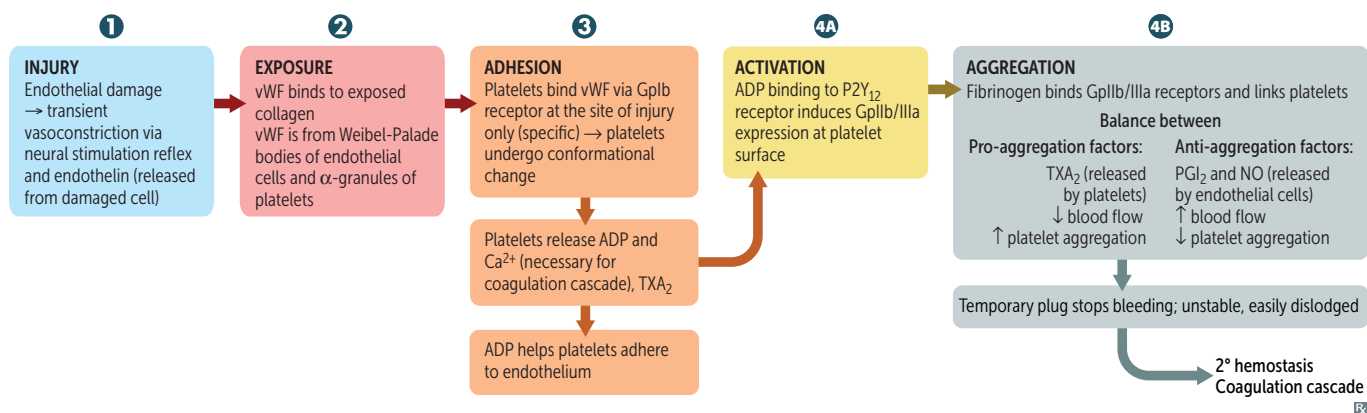
Heparin enhances the activity of antithrombin.

Principal targets of antithrombin: thrombin and factor Xa.

Factor V Leiden mutation produces a factor V resistant to inhibition by activated protein C.

tPA is used clinically as a thrombolytic.

Platelet plug formation (primary hemostasis)



Thrombogenesis

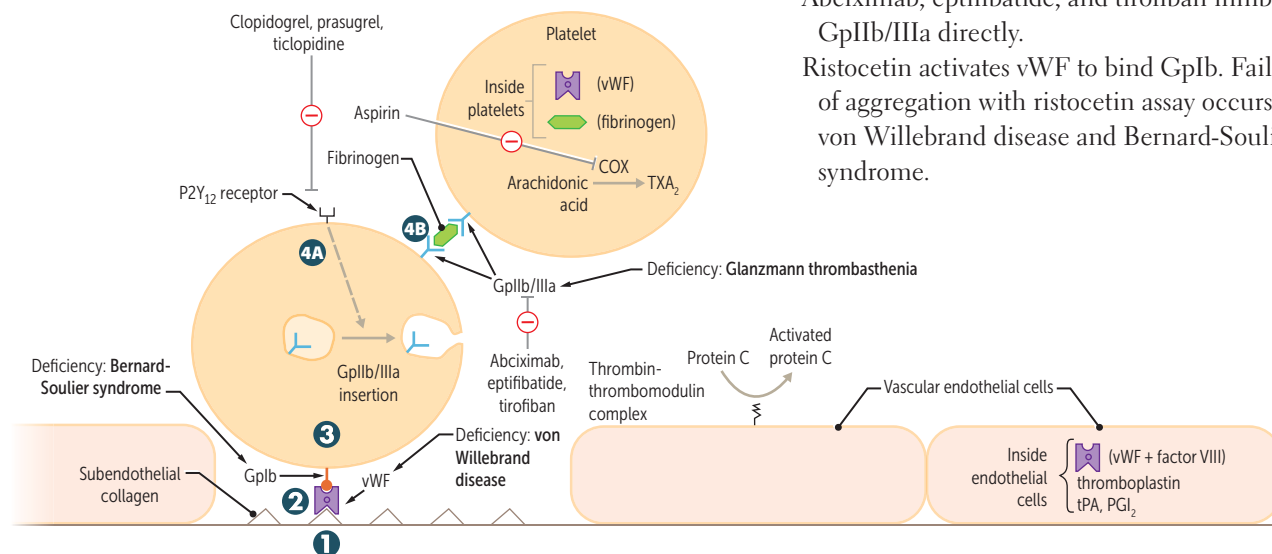
Formation of insoluble fibrin mesh.

Aspirin irreversibly inhibits cyclooxygenase, thereby inhibiting TXA_2 synthesis.

Clopidogrel, prasugrel, and ticlopidine inhibit ADP-induced expression of GpIIb/IIIa via P2Y_{12} receptor.

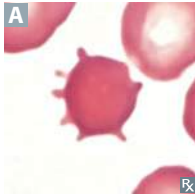
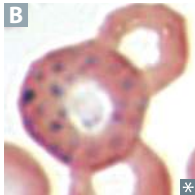
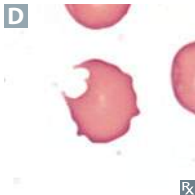
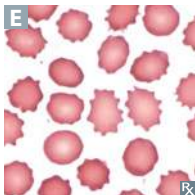
Abciximab, eptifibatide, and tirofiban inhibit GpIIb/IIIa directly.

Ristocetin activates vWF to bind GpIb. Failure of aggregation with ristocetin assay occurs in von Willebrand disease and Bernard-Soulier syndrome.

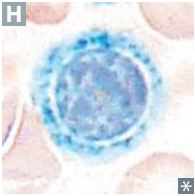
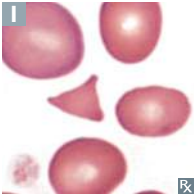
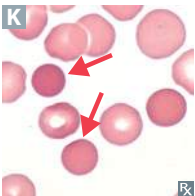
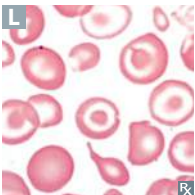


▶ HEMATOLOGY AND ONCOLOGY—PATHOLOGY

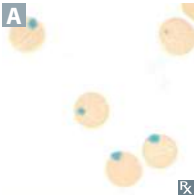
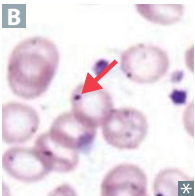
Pathologic RBC forms

TYPE	EXAMPLE	ASSOCIATED PATHOLOGY	NOTES
Acanthocyte ("spur cell") A		Liver disease, abetalipoproteinemia (states of cholesterol dysregulation).	<i>Acantho</i> = spiny.
Basophilic stippling B		Lead poisoning, sideroblastic anemias, myelodysplastic syndromes.	Seen primarily in peripheral smear, vs ringed sideroblasts seen in bone marrow. Aggregation of residual ribosomes.
Dacrocyte ("teardrop cell") C		Bone marrow infiltration (eg, myelofibrosis).	RBC "sheds a tear " because it's mechanically squeezed out of its home in the bone marrow.
Degmacyte ("bite cell") D		G6PD deficiency.	
Echinocyte ("burr cell") E		End-stage renal disease, liver disease, pyruvate kinase deficiency.	Different from acanthocyte; its projections are more uniform and smaller.
Elliptocyte F		Hereditary elliptocytosis, usually asymptomatic; caused by mutation in genes encoding RBC membrane proteins (eg, spectrin).	
Macro-ovalocyte G		Megaloblastic anemia (also hypersegmented PMNs).	

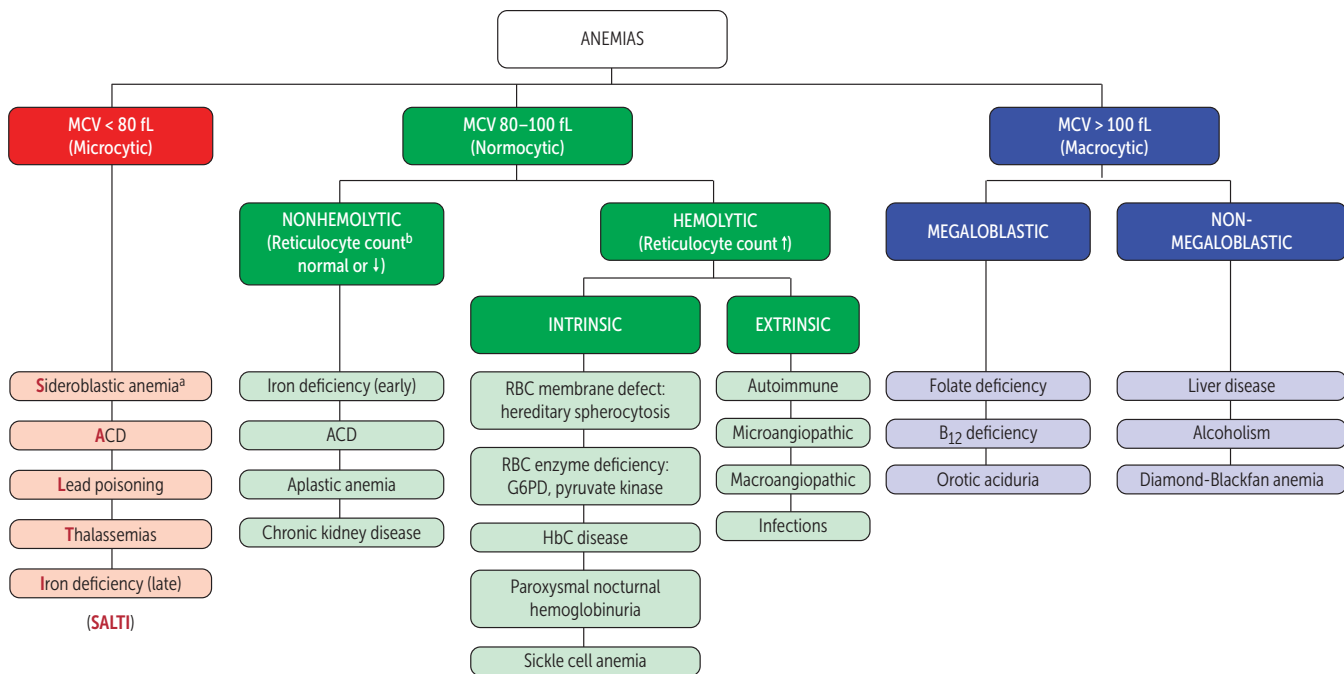
Pathologic RBC forms (continued)

TYPE	EXAMPLE	ASSOCIATED PATHOLOGY	NOTES
Ringed sideroblast H		Sideroblastic anemia. Excess iron in mitochondria.	Seen in bone marrow, vs basophilic stippling in peripheral smear.
Schistocyte I		Microangiopathic hemolytic anemias, including DIC, TTP/HUS, HELLP syndrome, mechanical hemolysis (eg, heart valve prosthesis).	Fragmented RBCs. Examples include helmet cell.
Sickle cell J		Sickle cell anemia.	Sickling occurs with dehydration, deoxygenation, and at high altitude.
Spherocyte K		Hereditary spherocytosis, drug- and infection-induced hemolytic anemia.	
Target cell L		HbC disease, Asplenia, Liver disease, Thalassemia.	“ HALT ,” said the hunter to his target .

Other RBC abnormalities

TYPE	EXAMPLE	ASSOCIATED PATHOLOGY	NOTES
Heinz bodies A		Seen in G6PD deficiency.	Oxidation of Hb -SH groups to -S-S- → Hb precipitation (Heinz bodies), with subsequent phagocytic damage to RBC membrane → bite cells.
Howell-Jolly bodies B		Seen in patients with functional hyposplenia or asplenia.	Basophilic nuclear remnants found in RBCs. Howell-Jolly bodies are normally removed from RBCs by splenic macrophages.

Anemias



On a peripheral blood smear, a lymphocyte nucleus is approximately the same size as a normocytic RBC. If RBC is larger than lymphocyte nucleus, consider macrocytosis; if RBC is smaller, consider microcytosis.

^aCopper deficiency can cause a microcytic sideroblastic anemia.

^bCorrected reticulocyte count (% reticulocytes × [patient hematocrit/normal hematocrit]) is used to determine if bone marrow response is adequate (> 2%).

Microcytic (MCV < 80 fL), hypochromic anemia

Iron deficiency

↓ iron due to chronic bleeding (eg, GI loss, menorrhagia), malnutrition, absorption disorders, GI surgery (eg, gastrectomy), or ↑ demand (eg, pregnancy) → ↓ final step in heme synthesis.

Labs: ↓ iron, ↑ TIBC, ↓ ferritin, ↑ free erythrocyte protoporphyrin. Microcytosis and hypochromasia (central pallor) **A**.

Symptoms: fatigue, conjunctival pallor **B**, pica (consumption of nonfood substances), spoon nails (koilonychia).

May manifest as glossitis, cheilosis, **Plummer-Vinson syndrome** (triad of iron deficiency anemia, esophageal webs, and dysphagia).

α-thalassemia

Defect: α-globin gene deletions → ↓ α-globin synthesis. *cis* deletion (both deletions occur on same chromosome) prevalent in Asian populations; *trans* deletion (deletions occur on separate chromosomes) prevalent in African populations.

4 allele deletion: no α-globin. Excess γ-globin forms γ₄ (Hb Barts). Incompatible with life (causes hydrops fetalis).

3 allele deletion: inheritance of chromosome with *cis* deletion + a chromosome with 1 allele deleted → HbH disease. Very little α-globin. Excess β-globin forms β₄ (HbH).

2 allele deletion: less clinically severe anemia.

1 allele deletion: no anemia (clinically silent).

Microcytic (MCV < 80 fL), hypochromic anemia (continued)**β-thalassemia**

Point mutations in splice sites and promoter sequences → ↓ β-globin synthesis. Prevalent in Mediterranean populations.

β-thalassemia minor (heterozygote): β chain is underproduced. Usually asymptomatic. Diagnosis confirmed by ↑ HbA₂ (> 3.5%) on electrophoresis.

β-thalassemia major (homozygote): β chain is absent → severe microcytic, hypochromic anemia with target cells and increased anisopoikilocytosis **C** requiring blood transfusion (2° hemochromatosis). Marrow expansion (“crew cut” on skull x-ray) → skeletal deformities. “Chipmunk” facies. Extramedullary hematopoiesis → hepatosplenomegaly. ↑ risk of parvovirus B19–induced aplastic crisis. ↑ HbF (α₂γ₂). HbF is protective in the infant and disease becomes symptomatic only after 6 months, when fetal hemoglobin declines.

HbS/β-thalassemia heterozygote: mild to moderate sickle cell disease depending on amount of β-globin production.

Lead poisoning

Lead inhibits ferrochelatase and ALA dehydratase → ↓ heme synthesis and ↑ RBC protoporphyrin. Also inhibits rRNA degradation → RBCs retain aggregates of rRNA (basophilic stippling).

Symptoms of **LEAD** poisoning:

- **L**ead **L**ines on gingivae (Burton lines) and on metaphyses of long bones **D** on x-ray.
- **E**ncephalopathy and **E**rythrocyte basophilic stippling.
- **A**bdominal colic and sideroblastic **A**nemia.
- **D**rops—wrist and foot drop. **D**imercaprol and **E**DTA are 1st line of treatment.

Succimer used for chelation for kids (It “sucks” to be a kid who eats lead).

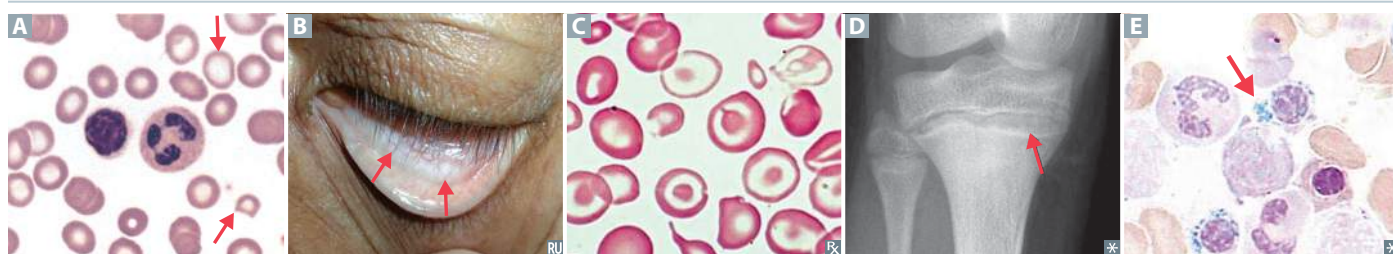
Exposure risk ↑ in old houses with chipped paint.

Sideroblastic anemia

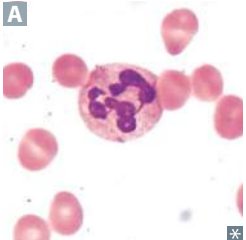
Causes: genetic (eg, X-linked defect in ALA synthase gene), acquired (myelodysplastic syndromes), and reversible (alcohol is most common; also lead, vitamin B₆ deficiency, copper deficiency, isoniazid).

Lab findings: ↑ iron, normal/↓ TIBC, ↑ ferritin. Ringed sideroblasts (with iron-laden, Prussian blue–stained mitochondria) seen in bone marrow **E**. Peripheral blood smear: basophilic stippling of RBCs.

Treatment: pyridoxine (B₆, cofactor for ALA synthase).



Macrocytic (MCV > 100 fL) anemia

	DESCRIPTION	FINDINGS
Megaloblastic anemia 	Impaired DNA synthesis → maturation of nucleus of precursor cells in bone marrow delayed relative to maturation of cytoplasm.	RBC macrocytosis, hypersegmented neutrophils A , glossitis.
Folate deficiency	Causes: malnutrition (eg, alcoholics), malabsorption, drugs (eg, methotrexate, trimethoprim, phenytoin), ↑ requirement (eg, hemolytic anemia, pregnancy).	↑ homocysteine, normal methylmalonic acid. No neurologic symptoms (vs B ₁₂ deficiency).
Vitamin B₁₂ (cobalamin) deficiency	Causes: insufficient intake (eg, veganism), malabsorption (eg, Crohn disease), pernicious anemia, <i>Diphyllobothrium latum</i> (fish tapeworm), gastrectomy.	↑ homocysteine, ↑ methylmalonic acid. Neurologic symptoms: reversible dementia, subacute combined degeneration (due to involvement of B ₁₂ in fatty acid pathways and myelin synthesis): spinocerebellar tract, lateral corticospinal tract, dorsal column dysfunction. Historically diagnosed with the Schilling test, a 4-stage test that determines if the cause is dietary insufficiency vs malabsorption. Anemia 2° to insufficient intake may take several years to develop due to liver's ability to store B ₁₂ (as opposed to folate deficiency).
Orotic aciduria	Inability to convert orotic acid to UMP (de novo pyrimidine synthesis pathway) because of defect in UMP synthase. Autosomal recessive. Presents in children as failure to thrive, developmental delay, and megaloblastic anemia refractory to folate and B ₁₂ . No hyperammonemia (vs ornithine transcarbamylase deficiency—↑ orotic acid with hyperammonemia).	Orotic acid in urine. Treatment: uridine monophosphate to bypass mutated enzyme.
Nonmegaloblastic anemia	Macrocytic anemia in which DNA synthesis is unimpaired. Causes: alcoholism, liver disease.	RBC macrocytosis without hypersegmented neutrophils.
Diamond-Blackfan anemia	Rapid-onset anemia within 1st year of life due to intrinsic defect in erythroid progenitor cells.	↑ % HbF (but ↓ total Hb). Short stature, craniofacial abnormalities, and upper extremity malformations (triphalangeal thumbs) in up to 50% of cases.

Normocytic, normochromic anemia

Normocytic, normochromic anemias are classified as nonhemolytic or hemolytic. The hemolytic anemias are further classified according to the cause of the hemolysis (intrinsic vs extrinsic to the RBC) and by the location of the hemolysis (intravascular vs extravascular). Hemolysis can lead to increases in LDH, reticulocytes, unconjugated bilirubin, urobilinogen in urine.

Intravascular hemolysis

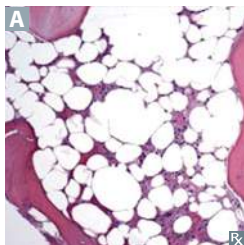
Findings: ↓ haptoglobin, ↑ schistocytes on blood smear. Characteristic hemoglobinuria, hemosiderinuria, and urobilinogen in urine. May also see ↑ unconjugated bilirubin. Notable causes are mechanical hemolysis (eg, prosthetic valve), paroxysmal nocturnal hemoglobinuria, microangiopathic hemolytic anemias.

Extravascular hemolysis

Findings: macrophages in spleen clear RBCs. Spherocytes in peripheral smear (most commonly hereditary spherocytosis and autoimmune hemolytic anemia), no hemoglobinuria/hemosiderinuria. Can present with urobilinogen in urine.

Nonhemolytic, normocytic anemia

	DESCRIPTION	FINDINGS
Anemia of chronic disease	Inflammation → ↑ hepcidin (released by liver, binds ferroportin on intestinal mucosal cells and macrophages, thus inhibiting iron transport) → ↓ release of iron from macrophages and ↓ iron absorption from gut. Associated with conditions such as rheumatoid arthritis, SLE, neoplastic disorders, and chronic kidney disease.	↓ iron, ↓ TIBC, ↑ ferritin. Normocytic, but can become microcytic. Treatment: address underlying cause of inflammation, judicious use of blood transfusion, consider erythropoiesis-stimulating agents (ESAs) such as EPO (chronic kidney disease only).
Aplastic anemia	Caused by failure or destruction of myeloid stem cells due to: ▪ Radiation and drugs (benzene, chloramphenicol, alkylating agents, antimetabolites) ▪ Viral agents (parvovirus B19, EBV, HIV, hepatitis viruses) ▪ Fanconi anemia (DNA repair defect causing bone marrow failure; macrocytosis may be seen on CBC); also short stature, ↑ incidence of tumors/leukemia, café-au-lait spots, thumb/radial defects ▪ Idiopathic (immune mediated, 1° stem cell defect); may follow acute hepatitis	↓ reticulocyte count, ↑ EPO. Pancytopenia characterized by severe anemia, leukopenia, and thrombocytopenia. Normal cell morphology, but hypocellular bone marrow with fatty infiltration A (dry bone marrow tap). Symptoms: fatigue, malaise, pallor, purpura, mucosal bleeding, petechiae, infection. Treatment: withdrawal of offending agent, immunosuppressive regimens (eg, antithymocyte globulin, cyclosporine), bone marrow allograft, RBC/platelet transfusion, bone marrow stimulation (eg, GM-CSF).

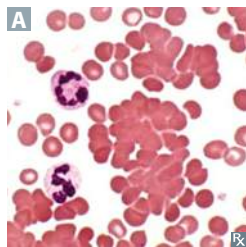


Intrinsic hemolytic anemia

	DESCRIPTION	FINDINGS
Hereditary spherocytosis	Extravascular hemolysis due to defect in proteins interacting with RBC membrane skeleton and plasma membrane (eg, ankyrin, band 3, protein 4.2, spectrin). Mostly autosomal dominant inheritance. Results in small, round RBCs with less surface area and no central pallor (↑ MCHC) → premature removal by spleen.	Splenomegaly, aplastic crisis (parvovirus B19 infection). Labs: osmotic fragility test ⊕. Normal to ↓ MCV with abundance of cells. Treatment: splenectomy.
G6PD deficiency	Most common enzymatic disorder of RBCs. Causes extravascular and intravascular hemolysis. X-linked recessive. Defect in G6PD → ↓ glutathione → ↑ RBC susceptibility to oxidant stress. Hemolytic anemia following oxidant stress (eg, sulfa drugs, antimalarials, infections, fava beans).	Back pain, hemoglobinuria a few days after oxidant stress . Labs: blood smear shows RBCs with Heinz bodies and bite cells. “ Stress makes me eat bites of fava beans with Heinz ketchup.”
Pyruvate kinase deficiency	Autosomal recessive pyruvate kinase defect → ↓ ATP → rigid RBCs → extravascular hemolysis. Increases levels of 2,3-BPG → ↓ hemoglobin affinity for O ₂ .	Hemolytic anemia in a newborn.
Paroxysmal nocturnal hemoglobinuria	↑ complement-mediated intravascular RBC lysis (impaired synthesis of GPI anchor for decay-accelerating factor that protects RBC membrane from complement). Acquired mutation in a hematopoietic stem cell. ↑ incidence of acute leukemias. Patients may report red or pink urine (from hemoglobinuria).	Associated with aplastic anemia. Triad: Coombs ⊖ hemolytic anemia, pancytopenia, and venous thrombosis. Labs: CD55/59 ⊖ RBCs on flow cytometry. Treatment: eculizumab (terminal complement inhibitor).
Sickle cell anemia	HbS point mutation causes a single amino acid replacement in β chain (substitution of glutamic acid with valine). Causes extravascular and intravascular hemolysis. Pathogenesis: low O ₂ , high altitude, or acidosis precipitates sickling (deoxygenated HbS polymerizes) → anemia, vaso-occlusive disease. Newborns are initially asymptomatic because of ↑ HbF and ↓ HbS.  A B	Complications in sickle cell disease: ▪ Aplastic crisis (due to parvovirus B19). ▪ Autosplenectomy (Howell-Jolly bodies) → ↑ risk of infection by encapsulated organisms (eg, <i>S pneumoniae</i>). ▪ Splenic infarct/sequestration crisis. ▪ <i>Salmonella</i> osteomyelitis. ▪ Painful crises (vaso-occlusive): dactylitis B (painful swelling of hands/feet), priapism, acute chest syndrome, avascular necrosis, stroke. ▪ Renal papillary necrosis (↓ Po₂ in papilla) and microhematuria (medullary infarcts). Diagnosis: hemoglobin electrophoresis. Treatment: hydroxyurea (↑ HbF), hydration.
HbC disease	Glutamic acid-to-lyCine (lysine) mutation in β-globin. Causes extravascular hemolysis.	Patients with HbSC (1 of each mutant gene) have milder disease than HbSS patients. Blood smear in homozygotes: hemoglobin C crystals inside RBCs, target cells.

Extrinsic hemolytic anemia

Autoimmune hemolytic anemia



DESCRIPTION

Warm (IgG)—chronic anemia seen in SLE and CLL and with certain drugs (eg, α -methylidopa) (“**w**arm **w**eather is **G**reat”).

Cold (IgM and complement)—acute anemia triggered by cold; seen in CLL, *Mycoplasma pneumoniae* infections, and infectious **M**ononucleosis (“**c**old weather is **M**MMiserable”). RBC agglutinates **A** may cause painful, blue fingers and toes with cold exposure.

Many warm and cold AIHAs are idiopathic in etiology.

FINDINGS

Autoimmune hemolytic anemias are usually Coombs $\oplus$.

Direct Coombs test—anti-Ig antibody (Coombs reagent) added to patient’s RBCs. RBCs agglutinate if RBCs are coated with Ig.

Indirect Coombs test—normal RBCs added to patient’s serum. If serum has anti-RBC surface Ig, RBCs agglutinate when Coombs reagent added.

	Patient component	Reagent(s)	$\oplus$ Result (agglutination)	$\ominus$ Result (no agglutination)
Direct Coombs	 RBCs +/- anti-RBC Ab	 Anti-human globulin (Coombs reagent)	 $\oplus$ Result Anti-RBC Ab present	 $\ominus$ Result Anti-RBC Ab absent
Indirect Coombs	 Patient serum +/- anti-donor RBC Ab	 Donor blood	 $\oplus$ Result Anti-donor RBC Ab present	 $\ominus$ Result Anti-donor RBC Ab absent
		 Anti-human globulin (Coombs reagent)		

Microangiopathic anemia

Pathogenesis: RBCs are damaged when passing through obstructed or narrowed vessel lumina. Seen in DIC, TTP/HUS, SLE, HELLP syndrome, and malignant hypertension.

Schistocytes (eg, “helmet cells”) are seen on peripheral blood smear due to mechanical destruction (*schisto* = to split) of RBCs.

Macroangiopathic anemia

Prosthetic heart valves and aortic stenosis may also cause hemolytic anemia 2° to mechanical destruction of RBCs.

Schistocytes on peripheral blood smear.

Infections

$\uparrow$ destruction of RBCs (eg, malaria, *Babesia*).

Lab values in anemia

	Iron deficiency	Chronic disease	Hemo-chromatosis	Pregnancy/OCP use
Serum iron	↓	↓	↑	—
Transferrin or TIBC	↑	↓ ^a	↓	↑
Ferritin	↓	↑	↑	—
% transferrin saturation (serum iron/TIBC)	↓↓	—	↑↑	↓

↑↓ = 1° disturbance.
Transferrin—**transports** iron in blood.
 TIBC—indirectly measures transferrin.
 Ferritin—1° iron storage protein of body.
^aEvolutionary reasoning—pathogens use circulating iron to thrive. The body has adapted a system in which iron is stored within the cells of the body and prevents pathogens from acquiring circulating iron.

Leukopenias

CELL TYPE	CELL COUNT	CAUSES
Neutropenia	Absolute neutrophil count < 1500 cells/mm ³ . Severe infections typical when < 500 cells/mm ³ .	Sepsis/postinfection, drugs (including chemotherapy), aplastic anemia, SLE, radiation
Lymphopenia	Absolute lymphocyte count < 1500 cells/mm ³ (< 3000 cells/mm ³ in children)	HIV, DiGeorge syndrome, SCID, SLE, corticosteroids, ^a radiation, sepsis, postoperative
Eosinopenia	Absolute eosinophil count < 30 cells/mm ³	Cushing syndrome, corticosteroids ^a

^aCorticosteroids cause neutrophilia, despite causing eosinopenia and lymphopenia. Corticosteroids ↓ activation of neutrophil adhesion molecules, impairing migration out of the vasculature to sites of inflammation. In contrast, corticosteroids sequester eosinophils in lymph nodes and cause apoptosis of lymphocytes.

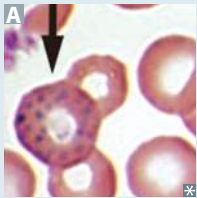
Left shift

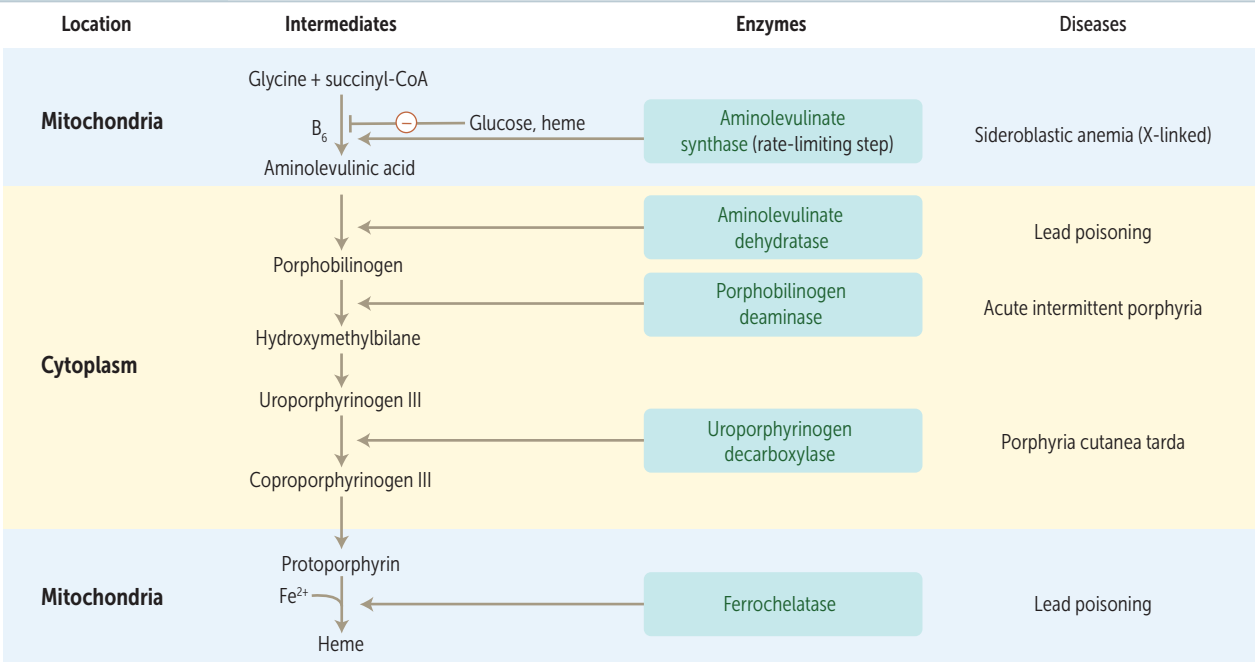
↑ neutrophil precursors, such as band cells and metamyelocytes, in peripheral blood. Usually seen with neutrophilia in the acute response to infection or inflammation. Called **leukoerythroblastic reaction** when left shift is seen with immature RBCs. Occurs with severe anemia (physiologic response) or marrow response (eg, fibrosis, tumor taking up space in marrow).

A **left shift** is a shift to a more **immature** cell in the maturation process.

Heme synthesis, porphyrias, and lead poisoning

The porphyrias are hereditary or acquired conditions of defective heme synthesis that lead to the accumulation of heme precursors. Lead inhibits specific enzymes needed in heme synthesis, leading to a similar condition.

CONDITION	AFFECTED ENZYME	ACCUMULATED SUBSTRATE	PRESENTING SYMPTOMS
Lead poisoning 	Ferrochelatase and ALA dehydratase	Protoporphyrin, ALA (blood)	Microcytic anemia (basophilic stippling in peripheral smear A , ringed sideroblasts in bone marrow), GI and kidney disease. Children—exposure to lead paint → mental deterioration. Adults—environmental exposure (eg, batteries, ammunition) → headache, memory loss, demyelination.
Acute intermittent porphyria	Porphobilinogen deaminase, previously known as uroporphyrinogen I synthase (autosomal dominant mutation)	Porphobilinogen, ALA	Symptoms (5 P's): ▪ Painful abdomen ▪ Port wine-colored urine ▪ Polyneuropathy ▪ Psychological disturbances ▪ Precipitated by drugs (eg, cytochrome P-450 inducers), alcohol, starvation Treatment: glucose and heme, which inhibit ALA synthase.
Porphyria cutanea tarda 	Uroporphyrinogen decarboxylase (autosomal dominant mutation)	Uroporphyrin (tea-colored urine)	Blistering cutaneous photosensitivity and hyperpigmentation B . Most common porphyria. Exacerbated with alcohol consumption.



↓ heme → ↑ ALA synthase activity
 ↑ heme → ↓ ALA synthase activity

Iron poisoning

High mortality rate with accidental ingestion by children (adult iron tablets may look like candy).

MECHANISM

Cell death due to peroxidation of membrane lipids.

SYMPTOMS/SIGNS

Nausea, vomiting, gastric bleeding, lethargy, scarring leading to GI obstruction.

TREATMENT

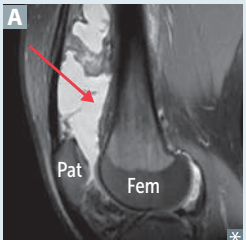
Chelation (eg, IV deferoxamine, oral deferasirox) and dialysis.

Coagulation disorders

PT—tests function of common and extrinsic pathway (factors I, II, V, VII, and X). Defect → ↑ PT. INR (international normalized ratio)—calculated from PT. 1 = normal, > 1 = prolonged. Most common test used to follow patients on warfarin.

PTT—tests function of common and intrinsic pathway (all factors except VII and XIII). Defect → ↑ PTT.

Coagulation disorders can be due to clotting factor deficiencies or acquired inhibitors. Diagnosed with a mixing study, in which normal plasma is added to patient's plasma. Clotting factor deficiencies should correct (the PT or PTT returns to within the appropriate normal range), whereas factor inhibitors will not correct.

DISORDER	PT	PTT	MECHANISM AND COMMENTS
Hemophilia A, B, or C 	—	↑	Intrinsic pathway coagulation defect. ▪ A: deficiency of factor VIII → ↑ PTT; X-linked recessive. ▪ B: deficiency of factor IX → ↑ PTT; X-linked recessive. ▪ C: deficiency of factor XI → ↑ PTT; autosomal recessive. Macrohemorrhage in hemophilia—hemarthroses (bleeding into joints, such as knee A), easy bruising, bleeding after trauma or surgery (eg, dental procedures). Treatment: desmopressin + factor VIII concentrate (A); factor IX concentrate (B); factor XI concentrate (C).
Vitamin K deficiency	↑	↑	General coagulation defect. Bleeding time normal. ↓ activity of factors II, VII, IX, X, protein C, protein S.

Platelet disorders

Defects in platelet plug formation → ↑ bleeding time (BT).

Platelet abnormalities → microhemorrhage: mucous membrane bleeding, epistaxis, petechiae, purpura, ↑ bleeding time, possibly decreased platelet count (PC).

DISORDER	PC	BT	MECHANISM AND COMMENTS
Bernard-Soulier syndrome	–/↓	↑	Defect in platelet plug formation. Large platelets. ↓ GpIb → defect in platelet-to-vWF adhesion.
Glanzmann thrombasthenia	–	↑	Defect in platelet integrin $\alpha_{IIb}\beta_3$ (GpIIb/IIIa) → defect in platelet-to-platelet aggregation, and therefore platelet plug formation. Labs: blood smear shows no platelet clumping.
Hemolytic-uremic syndrome	↓	↑	Characterized by thrombocytopenia, microangiopathic hemolytic anemia, and acute renal failure. Typical HUS is seen in children, accompanied by diarrhea and commonly caused by enterohemorrhagic <i>E coli</i> (EHEC) (eg, O157:H7). HUS in adults does not present with diarrhea; EHEC infection not required. Same spectrum as TTP, with a similar clinical presentation and same initial treatment of plasmapheresis.
Immune thrombocytopenia	↓	↑	Anti-GpIIb/IIIa antibodies → splenic macrophage consumption of platelet-antibody complex. May be 1° (idiopathic) or 2° to autoimmune disorder, viral illness, malignancy, or drug reaction. Labs: ↑ megakaryocytes on bone marrow biopsy. Treatment: steroids, IVIG, splenectomy (for refractory ITP).
Thrombotic thrombocytopenic purpura	↓	↑	Inhibition or deficiency of ADAMTS 13 (vWF metalloprotease) → ↓ degradation of vWF multimers. Pathogenesis: ↑ large vWF multimers → ↑ platelet adhesion → ↑ platelet aggregation and thrombosis. Labs: schistocytes, ↑ LDH, normal coagulation parameters. Symptoms: pentad of neurologic and renal symptoms, fever, thrombocytopenia, and microangiopathic hemolytic anemia. Treatment: plasmapheresis, steroids.

Mixed platelet and coagulation disorders

DISORDER	PC	BT	PT	PTT	MECHANISM AND COMMENTS
von Willebrand disease	—	↑	—	—/↑	Intrinsic pathway coagulation defect: ↓ vWF → ↑ PTT (vWF acts to carry/protect factor VIII). Defect in platelet plug formation: ↓ vWF → defect in platelet-to-vWF adhesion. Autosomal dominant. Mild but most common inherited bleeding disorder. No platelet aggregation with ristocetin cofactor assay. Treatment: desmopressin, which releases vWF stored in endothelium.
Disseminated intravascular coagulation	↓	↑	↑	↑	Widespread activation of clotting → deficiency in clotting factors → bleeding state. Causes: Sepsis (gram ⊖), Trauma, Obstetric complications, acute Pancreatitis, Malignancy, Nephrotic syndrome, Transfusion (STOP Making New Thrombi). Labs: schistocytes, ↑ fibrin degradation products (D-dimers), ↓ fibrinogen, ↓ factors V and VIII.

Hereditary thrombosis syndromes leading to hypercoagulability

DISEASE	DESCRIPTION
Antithrombin deficiency	Inherited deficiency of antithrombin: has no direct effect on the PT, PTT, or thrombin time but diminishes the increase in PTT following heparin administration. Can also be acquired: renal failure/nephrotic syndrome → antithrombin loss in urine → ↓ inhibition of factors IIa and Xa.
Factor V Leiden	Production of mutant factor V (G → A DNA point mutation → Arg506Gln mutation near the cleavage site) that is resistant to degradation by activated protein C. Most common cause of inherited hypercoagulability in Caucasians. Complications include DVT, cerebral vein thromboses, recurrent pregnancy loss.
Protein C or S deficiency	↓ ability to inactivate factors Va and VIIIa. ↑ risk of thrombotic skin necrosis with hemorrhage after administration of warfarin. If this occurs, think protein C deficiency. Together, protein C Cancels, and protein S Stops, coagulation.
Prothrombin gene mutation	Mutation in 3′ untranslated region → ↑ production of prothrombin → ↑ plasma levels and venous clots.

Blood transfusion therapy

COMPONENT	DOSAGE EFFECT	CLINICAL USE
Packed RBCs	↑ Hb and O ₂ carrying capacity	Acute blood loss, severe anemia
Platelets	↑ platelet count (↑ ~ 5000/mm ³ /unit)	Stop significant bleeding (thrombocytopenia, qualitative platelet defects)
Fresh frozen plasma/ prothrombin complex concentrate	↑ coagulation factor levels	DIC, cirrhosis, immediate anticoagulation reversal
Cryoprecipitate	Contains fibrinogen, factor VIII, factor XIII, vWF, and fibronectin	Coagulation factor deficiencies involving fibrinogen and factor VIII

Blood transfusion risks include infection transmission (low), transfusion reactions, iron overload (may lead to 2° hemochromatosis), hypocalcemia (citrate is a Ca²⁺ chelator), and hyperkalemia (RBCs may lyse in old blood units).

Leukemia vs lymphoma

Leukemia	Lymphoid or myeloid neoplasm with widespread involvement of bone marrow. Tumor cells are usually found in peripheral blood.
Lymphoma	Discrete tumor mass arising from lymph nodes. Presentations often blur definitions.

Hodgkin vs non-Hodgkin lymphoma**Hodgkin****Non-Hodgkin**

Both may present with constitutional (“B”) signs/symptoms: low-grade fever, night sweats, weight loss (patients are **B**othered by **B** symptoms).

Localized, single group of nodes; contiguous spread (stage is strongest predictor of prognosis). Overall prognosis better than that of non-Hodgkin lymphoma.

Multiple lymph nodes involved; extranodal involvement common; noncontiguous spread.

Characterized by Reed-Sternberg cells.

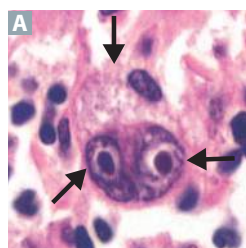
Majority involve B cells; a few are of T-cell lineage.

Bimodal distribution—young adulthood and > 55 years; more common in men except for nodular sclerosing type.

Can occur in children and adults.

Associated with EBV.

May be associated with HIV and autoimmune diseases.

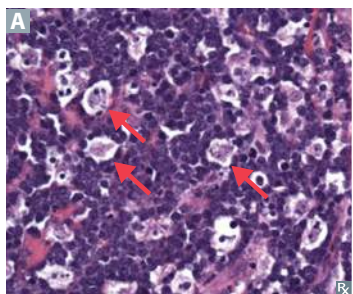
Hodgkin lymphoma

Contains Reed-Sternberg cells: distinctive tumor giant cells; binucleate or bilobed with the **2** halves as mirror images (“owl eyes” **A**). **2** owl eyes × **15** = **30**. RS cells are CD**15**⁺ and CD**30**⁺ B-cell origin. Necessary but not sufficient for a diagnosis of Hodgkin lymphoma.

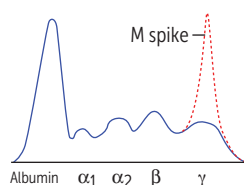
SUBTYPE	NOTES
Nodular sclerosis	Most common
Lymphocyte rich	Best prognosis
Mixed cellularity	Eosinophilia, seen in immunocompromised patients
Lymphocyte depleted	Seen in immunocompromised patients

Non-Hodgkin lymphoma

TYPE	OCCURS IN	GENETICS	COMMENTS
Neoplasms of mature B cells			
Burkitt lymphoma	Adolescents or young adults	t(8;14)—translocation of <i>c-myc</i> (8) and heavy-chain Ig (14)	“Starry sky” appearance, sheets of lymphocytes with interspersed “tingible body” macrophages (arrows in A). Associated with EBV. Jaw lesion B in endemic form in Africa; pelvis or abdomen in sporadic form.
Diffuse large B-cell lymphoma	Usually older adults, but 20% in children	Alterations in Bcl-2, Bcl-6	Most common type of non-Hodgkin lymphoma in adults.
Follicular lymphoma	Adults	t(14;18)—translocation of heavy-chain Ig (14) and <i>BCL-2</i> (18)	Indolent course; Bcl-2 inhibits apoptosis. Presents with painless “waxing and waning” lymphadenopathy. Follicular architecture: small cleaved cells (grade 1), large cells (grade 3), or mixture (grade 2).
Mantle cell lymphoma	Adult males	t(11;14)—translocation of cyclin D1 (11) and heavy-chain Ig (14), CD 5+	Very aggressive, patients typically present with late-stage disease.
Marginal zone lymphoma	Adults	t(11,18)	Associated with chronic inflammation (eg, Sjögren syndrome, chronic gastritis [MALT lymphoma]).
Primary central nervous system lymphoma	Adults	Most commonly associated with HIV/AIDS; pathogenesis involves EBV infection	Considered an AIDS-defining illness. Variable presentation: confusion, memory loss, seizures. Mass lesion(s) on MRI C , needs to be distinguished from toxoplasmosis via CSF analysis or other lab tests.
Neoplasms of mature T cells			
Adult T-cell lymphoma	Adults	Caused by HTLV (associated with IV drug abuse)	Adults present with cutaneous lesions; common in Japan, West Africa, and the Caribbean. Lytic bone lesions, hypercalcemia.
Mycosis fungoides/Sézary syndrome	Adults		Mycosis fungoides: skin patches D /plaques (cutaneous T-cell lymphoma), characterized by atypical CD4+ cells with “cerebriform” nuclei and intraepidermal neoplastic cell aggregates (Pautrier microabscess). May progress to Sézary syndrome (T-cell leukemia).



Multiple myeloma



Monoclonal plasma cell (“fried egg” appearance) cancer that arises in the marrow and produces large amounts of IgG (55%) or IgA (25%). Bone marrow > 10% monoclonal plasma cells. Most common 1° tumor arising within bone in people > 40–50 years old.

Associated with:

- ↑ susceptibility to infection
- Primary amyloidosis (AL)
- Punched-out lytic bone lesions on x-ray **A**
- M spike on serum protein electrophoresis
- Ig light chains in urine (Bence Jones protein)
- Rouleaux formation **B** (RBCs stacked like poker chips in blood smear)

Numerous plasma cells **C** with “clock-face” chromatin and intracytoplasmic inclusions containing immunoglobulin.

Monoclonal gammopathy of undetermined significance (MGUS)—monoclonal expansion of plasma cells (bone marrow < 10% monoclonal plasma cells), asymptomatic, may lead to multiple myeloma. No “**CRAB**” findings. Patients with MGUS develop multiple myeloma at a rate of 1–2% per year.

Think **CRAB**:

Hyper**C**alcemia
Renal involvement
Anemia

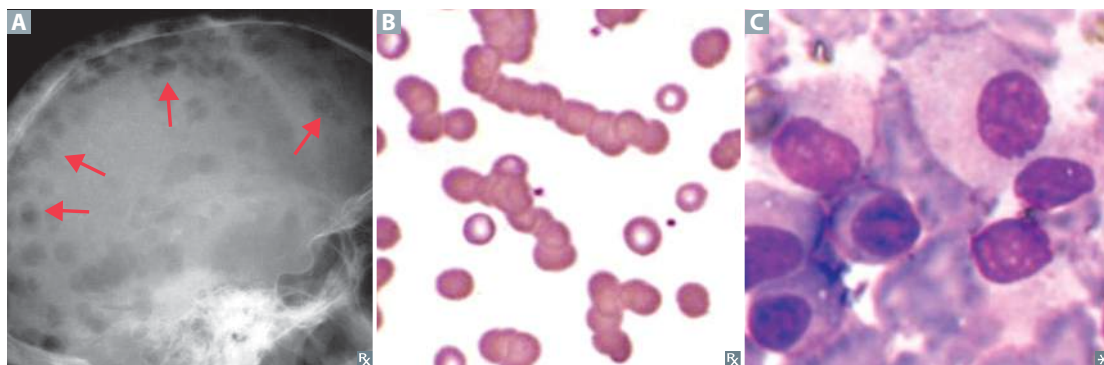
Bone lytic lesions/**B**ack pain

Multiple **M**yeloma: Monoclonal **M** protein spike

Distinguish from **Waldenström**

macroglobulinemia → M spike = IgM

→ hyperviscosity syndrome (eg, blurred vision, Raynaud phenomenon); no “**CRAB**” findings.



Myelodysplastic syndromes

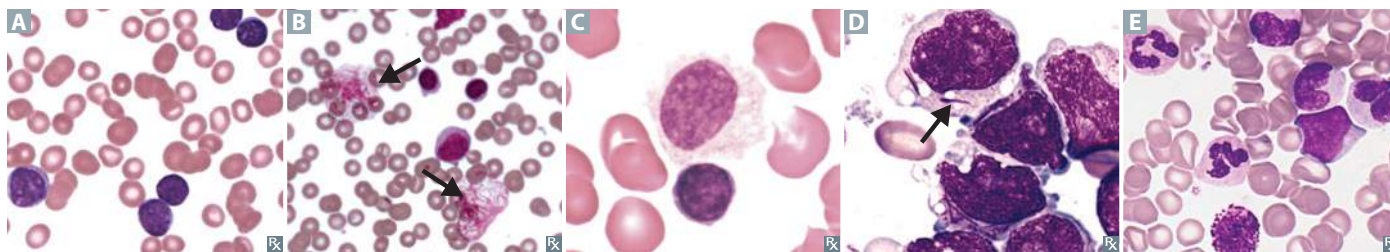
Stem-cell disorders involving ineffective hematopoiesis → defects in cell maturation of nonlymphoid lineages. Caused by de novo mutations or environmental exposure (eg, radiation, benzene, chemotherapy). Risk of transformation to AML.

Pseudo-Pelger-Huet anomaly—neutrophils with bilobed nuclei. Typically seen after chemotherapy.

Leukemias

Unregulated growth and differentiation of WBCs in bone marrow → marrow failure → anemia (↓ RBCs), infections (↓ mature WBCs), and hemorrhage (↓ platelets). Usually presents with ↑ circulating WBCs (malignant leukocytes in blood); rare cases present with normal/↓ WBCs. Leukemic cell infiltration of liver, spleen, lymph nodes, and skin (leukemia cutis) possible.

TYPE	NOTES
Lymphoid neoplasms	
Acute lymphoblastic leukemia/lymphoma	Most frequently occurs in children; less common in adults (worse prognosis). T-cell ALL can present as mediastinal mass (presenting as SVC-like syndrome). Associated with Down syndrome. Peripheral blood and bone marrow have ↑↑↑ lymphoblasts A. TdT+ (marker of pre-T and pre-B cells), CD10+ (marker of pre-B cells). Most responsive to therapy. May spread to CNS and testes. t(12;21) → better prognosis.
Chronic lymphocytic leukemia/small lymphocytic lymphoma	Age: > 60 years. Most common adult leukemia. CD20+, CD23+, CD5+ B-cell neoplasm. Often asymptomatic, progresses slowly; smudge cells B in peripheral blood smear; autoimmune hemolytic anemia. CLL = Crushed Little Lymphocytes (smudge cells). Richter transformation—CLL/SLL transformation into an aggressive lymphoma, most commonly diffuse large B-cell lymphoma (DLBCL).
Hairy cell leukemia	Age: Adult males. Mature B-cell tumor. Cells have filamentous, hair-like projections (fuzzy appearing on LM C). Peripheral lymphadenopathy is uncommon. Causes marrow fibrosis → dry tap on aspiration. Patients usually present with massive splenomegaly and pancytopenia. Stains TRAP (tartrate-resistant acid phosphatase) ⊕. TRAP stain largely replaced with flow cytometry. Treatment: cladribine, pentostatin.
Myeloid neoplasms	
Acute myelogenous leukemia	Median onset 65 years. Auer rods D; myeloperoxidase ⊕ cytoplasmic inclusions seen mostly in APL (formerly M3 AML); ↑↑↑ circulating myeloblasts on peripheral smear; adults. Risk factors: prior exposure to alkylating chemotherapy, radiation, myeloproliferative disorders, Down syndrome. APL: t(15;17), responds to all-<i>trans</i> retinoic acid (vitamin A), inducing differentiation of promyelocytes; DIC is a common presentation.
Chronic myelogenous leukemia	Occurs across the age spectrum with peak incidence 45–85 years, median age at diagnosis 64 years. Defined by the Philadelphia chromosome (t[9;22], <i>BCR-ABL</i>) and myeloid stem cell proliferation. Presents with dysregulated production of mature and maturing granulocytes (eg, neutrophils, metamyelocytes, myelocytes, basophils E) and splenomegaly. May accelerate and transform to AML or ALL (“blast crisis”). Very low LAP as a result of low activity in malignant neutrophils (vs benign neutrophilia [leukemoid reaction], in which LAP is ↑). Responds to <i>bcr-abl</i> tyrosine kinase inhibitors (eg, imatinib).



Chronic myeloproliferative disorders

The myeloproliferative disorders (polycythemia vera, essential thrombocythemia, myelofibrosis, and CML) are malignant hematopoietic neoplasms with varying impacts on WBCs and myeloid cell lines. Associated with V617F JAK2 mutation.

Polycythemia vera

Primary polycythemia. Disorder of ↑ RBCs. May present as intense itching after hot shower. Rare but classic symptom is erythromelalgia (severe, burning pain and red-blue coloration) due to episodic blood clots in vessels of the extremities **A**.

↓ EPO (vs 2° polycythemia, which presents with endogenous or artificially ↑ EPO).

Treatment: phlebotomy, hydroxyurea, ruxolitinib (JAK1/2 inhibitor).

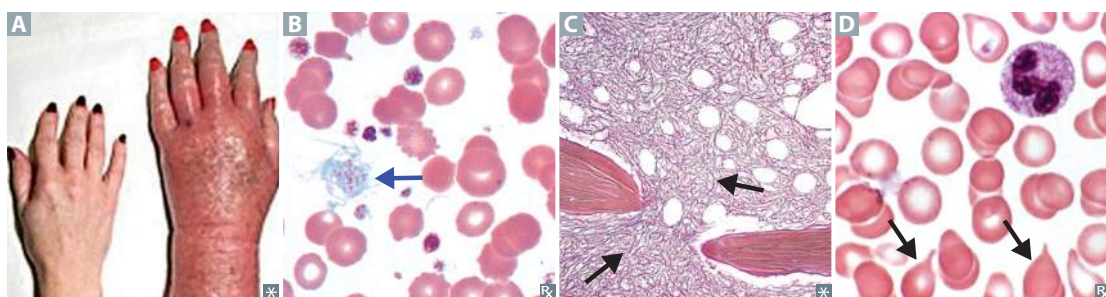
Essential thrombocythemia

Characterized by massive proliferation of megakaryocytes and platelets. Symptoms include bleeding and thrombosis. Blood smear shows markedly increased number of platelets, which may be large or otherwise abnormally formed **B**. Erythromelalgia may occur.

Myelofibrosis

Obliteration of bone marrow with fibrosis **C** due to ↑ fibroblast activity. Often associated with massive splenomegaly and “teardrop” RBCs **D**. “Bone marrow is **crying** because it’s fibrosed and is a dry tap.”

	RBCs	WBCs	PLATELETS	PHILADELPHIA CHROMOSOME	JAK2 MUTATIONS
Polycythemia vera	↑	↑	↑	⊖	⊕
Essential thrombocythemia	—	—	↑	⊖	⊕ (30–50%)
Myelofibrosis	↓	Variable	Variable	⊖	⊕ (30–50%)
CML	↓	↑	↑	⊕	⊖

**Polycythemia**

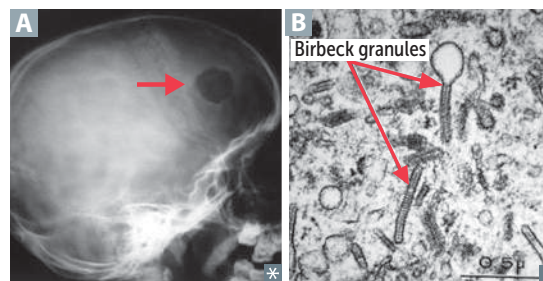
	PLASMA VOLUME	RBC MASS	O ₂ SATURATION	EPO LEVELS	ASSOCIATIONS
Relative	↓	—	—	—	Dehydration, burns.
Appropriate absolute	—	↑	↓	↑	Lung disease, congenital heart disease, high altitude.
Inappropriate absolute	—	↑	—	↑	Malignancy (eg, renal cell carcinoma, hepatocellular carcinoma), hydronephrosis. Due to ectopic EPO secretion.
Polycythemia vera	↑	↑↑	—	↓	EPO ↓ in PCV due to negative feedback suppressing renal EPO production.

Chromosomal translocations

TRANSLOCATION	ASSOCIATED DISORDER	
t(8;14)	Burkitt lymphoma (<i>c-myc</i> activation)	
t(9;22) (Philadelphia chromosome)	CML (<i>BCR-ABL</i> hybrid), ALL (less common, poor prognostic factor)	Philadelphia CreaML cheese. The Ig heavy chain genes on chromosome 14 are constitutively expressed. When other genes (eg, <i>c-myc</i> and <i>BCL-2</i>) are translocated next to this heavy chain gene region, they are overexpressed.
t(11;14)	Mantle cell lymphoma (cyclin D1 activation)	
t(14;18)	Follicular lymphoma (<i>BCL-2</i> activation)	
t(15;17)	APL (M3 type of AML)	Responds to all- <i>trans</i> retinoic acid.

Langerhans cell histiocytosis

Collective group of proliferative disorders of dendritic (Langerhans) cells. Presents in a child as lytic bone lesions **A** and skin rash or as recurrent otitis media with a mass involving the mastoid bone. Cells are functionally immature and do not effectively stimulate primary T cells via antigen presentation. Cells express S-100 (mesodermal origin) and CD1a. Birbeck granules (“tennis rackets” or rod shaped on EM) are characteristic **B**.



▶ HEMATOLOGY AND ONCOLOGY—PHARMACOLOGY

Heparin

MECHANISM	Lowers the activity of thrombin and factor Xa. Short half-life.
CLINICAL USE	Immediate anticoagulation for pulmonary embolism (PE), acute coronary syndrome, MI, deep venous thrombosis (DVT). Used during pregnancy (does not cross placenta). Follow PTT.
ADVERSE EFFECTS	Bleeding, thrombocytopenia (HIT), osteoporosis, drug-drug interactions. For rapid reversal (antidote), use protamine sulfate (positively charged molecule that binds negatively charged heparin).
NOTES	Low-molecular-weight heparins (eg, enoxaparin, dalteparin) act predominantly on factor Xa. Fondaparinux acts only on factor Xa. Have better bioavailability and 2–4× longer half life than unfractionated heparin; can have better bioavailability, and 2–4 times longer half life; can be administered subcutaneously and without laboratory monitoring. Not easily reversible. Heparin-induced thrombocytopenia (HIT)—development of IgG antibodies against heparin-bound platelet factor 4 (PF4). Antibody-heparin-PF4 complex activates platelets → thrombosis and thrombocytopenia.

Direct thrombin inhibitors

	Bivalirudin (related to hirudin, the anticoagulant used by leeches), argatroban, dabigatran (only oral agent in class).
MECHANISM	Directly inhibits activity of free and clot-associated thrombin.
CLINICAL USE	Venous thromboembolism, atrial fibrillation. Can be used in HIT. Does not require lab monitoring.
ADVERSE EFFECTS	Bleeding; can reverse dabigatran with idarucizumab. Consider PCC and/or antifibrinolytics (eg, tranexamic acid) if no reversal agent available.

Warfarin

MECHANISM	Interferes with γ -carboxylation of vitamin K–dependent clotting factors II, VII, IX, and X, and proteins C and S. Metabolism affected by polymorphisms in the gene for vitamin K epoxide reductase complex (VKORC1). In laboratory assay, has effect on EX trinsic pathway and $\uparrow$ PT . Long half-life.	The EX -President T went to war (farin).
CLINICAL USE	Chronic anticoagulation (eg, venous thromboembolism prophylaxis, and prevention of stroke in atrial fibrillation). Not used in pregnant women (because warfarin, unlike heparin, crosses placenta). Follow PT/INR.	
ADVERSE EFFECTS	Bleeding, teratogenic, skin/tissue necrosis A , drug-drug interactions. Proteins C and S have shorter half-lives than clotting factors II, VII, IX, and X, resulting in early transient hypercoagulability with warfarin use. Skin/tissue necrosis within first few days of large doses believed to be due to small vessel microthrombosis.	For reversal of warfarin, give vitamin K. For rapid reversal, give fresh frozen plasma (FFP) or PCC. Heparin “bridging”: heparin frequently used when starting warfarin. Heparin’s activation of antithrombin enables anticoagulation during initial, transient hypercoagulable state caused by warfarin. Initial heparin therapy reduces risk of recurrent venous thromboembolism and skin/tissue necrosis.



Heparin vs warfarin

	Heparin	Warfarin
ROUTE OF ADMINISTRATION	Parenteral (IV, SC)	Oral
SITE OF ACTION	Blood	Liver
ONSET OF ACTION	Rapid (seconds)	Slow, limited by half-lives of normal clotting factors
MECHANISM OF ACTION	Activates antithrombin, which $\downarrow$ the action of IIa (thrombin) and factor Xa	Impairs synthesis of vitamin K–dependent clotting factors II, VII, IX, and X, and anti-clotting proteins C and S
DURATION OF ACTION	Hours	Days
AGENTS FOR REVERSAL	Protamine sulfate	Vitamin K, FFP, PCC
MONITORING	PTT (intrinsic pathway)	PT/INR (extrinsic pathway)
CROSSES PLACENTA	No	Yes (teratogenic)

Direct factor Xa inhibitors

ApiXaban, rivaroXaban.

MECHANISM	Bind to and directly inhibit factor Xa.
CLINICAL USE	Treatment and prophylaxis for DVT and PE; stroke prophylaxis in patients with atrial fibrillation. Oral agents do not usually require coagulation monitoring.
ADVERSE EFFECTS	Bleeding. Not easily reversible.

Thrombolytics

Alteplase (tPA), reteplase (rPA), streptokinase, tenecteplase (TNK-tPA).

MECHANISM	Directly or indirectly aid conversion of plasminogen to plasmin, which cleaves thrombin and fibrin clots. ↑ PT, ↑ PTT, no change in platelet count.
CLINICAL USE	Early MI, early ischemic stroke, direct thrombolysis of severe PE.
ADVERSE EFFECTS	Bleeding. Contraindicated in patients with active bleeding, history of intracranial bleeding, recent surgery, known bleeding diatheses, or severe hypertension. Nonspecific reversal with antifibrinolytics (eg, aminocaproic acid, tranexamic acid), platelet transfusions, and factor corrections (eg, cryoprecipitate, FFP, PCC).

ADP receptor inhibitors

Clopidogrel, prasugrel, ticagrelor (reversible), ticlopidine.

MECHANISM	Inhibit platelet aggregation by irreversibly blocking ADP (P2Y ₁₂) receptor. Prevent expression of glycoproteins IIb/IIIa on platelet surface.
CLINICAL USE	Acute coronary syndrome; coronary stenting. ↓ incidence or recurrence of thrombotic stroke.
ADVERSE EFFECTS	Neutropenia (ticlopidine). TTP may be seen.

Cilostazol, dipyridamole

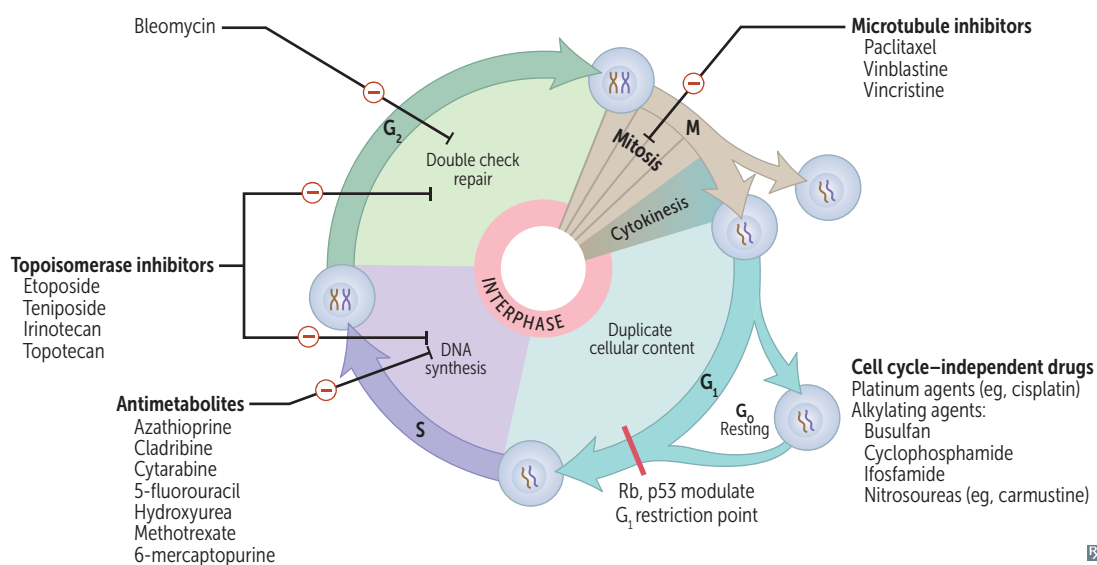
MECHANISM	Phosphodiesterase inhibitors; ↑ cAMP in platelets, resulting in inhibition of platelet aggregation; vasodilators.
CLINICAL USE	Intermittent claudication, coronary vasodilation, prevention of stroke or TIAs (combined with aspirin).
ADVERSE EFFECTS	Nausea, headache, facial flushing, hypotension, abdominal pain.

Glycoprotein IIb/IIIa inhibitors

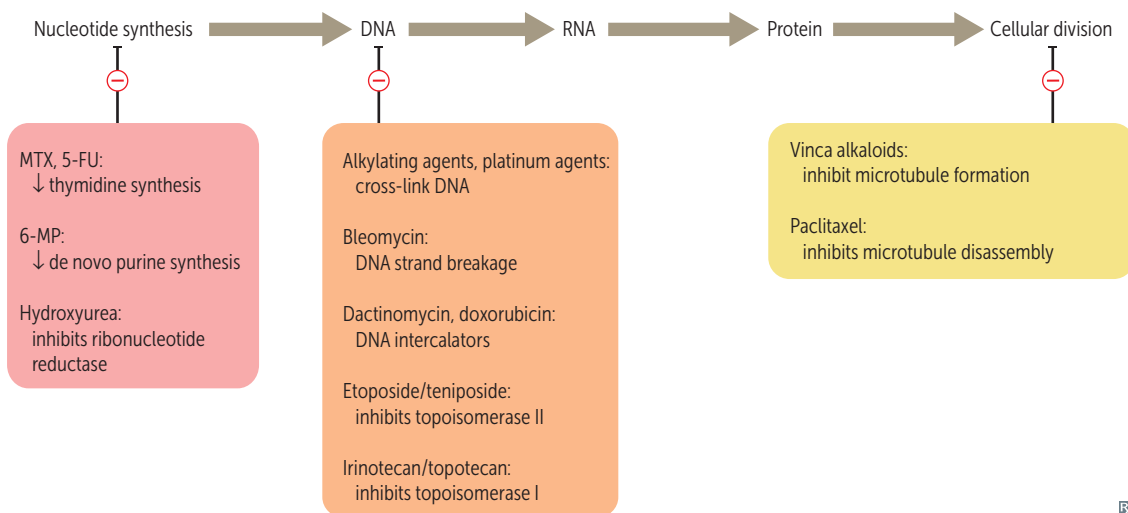
Abciximab, eptifibatide, tirofiban.

MECHANISM	Bind to the glycoprotein receptor IIb/IIIa on activated platelets, preventing aggregation. Abciximab is made from monoclonal antibody Fab fragments.
CLINICAL USE	Unstable angina, percutaneous coronary intervention.
ADVERSE EFFECTS	Bleeding, thrombocytopenia.

Cancer drugs—cell cycle



Cancer drugs—targets



Antimetabolites

DRUG	MECHANISM ^a	CLINICAL USE	ADVERSE EFFECTS
Azathioprine, 6-mercaptopurine	Purine (thiol) analogs → ↓ de novo purine synthesis. Activated by HGPRT. Azathioprine is metabolized into 6-MP.	Preventing organ rejection, rheumatoid arthritis, IBD, SLE; used to wean patients off steroids in chronic disease and to treat steroid-refractory chronic disease.	Myelosuppression, GI, liver. Azathioprine and 6-MP are metabolized by xanthine oxidase; thus both have ↑ toxicity with allopurinol or febuxostat.
Cladribine	Purine analog → multiple mechanisms (eg, inhibition of DNA polymerase, DNA strand breaks).	Hairy cell leukemia.	Myelosuppression, nephrotoxicity, and neurotoxicity.
Cytarabine (arabinofuranosyl cytidine)	Pyrimidine analog → inhibition of DNA polymerase.	Leukemias (AML), lymphomas.	Myelosuppression with megaloblastic anemia. CYT arabine causes pan CYT openia.
5-fluorouracil	Pyrimidine analog bioactivated to 5-FdUMP, which covalently complexes with thymidylate synthase and folic acid. Capecitabine is a prodrug with similar activity. This complex inhibits thymidylate synthase → ↓ dTMP → ↓ DNA synthesis.	Colon cancer, pancreatic cancer, basal cell carcinoma (topical). Effects enhanced with the addition of leucovorin.	Myelosuppression, palmar-plantar erythrodysesthesia (hand-foot syndrome).
Methotrexate	Folic acid analog that competitively inhibits dihydrofolate reductase → ↓ dTMP → ↓ DNA synthesis.	Cancers: leukemias (ALL), lymphomas, choriocarcinoma, sarcomas. Non-neoplastic: ectopic pregnancy, medical abortion (with misoprostol), rheumatoid arthritis, psoriasis, IBD, vasculitis.	Myelosuppression, which is reversible with leucovorin “rescue.” Hepatotoxicity. Mucositis (eg, mouth ulcers). Pulmonary fibrosis.

^aAll are S-phase specific.

Antitumor antibiotics

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Bleomycin	Induces free radical formation → breaks in DNA strands.	Testicular cancer, Hodgkin lymphoma.	Pulmonary fibrosis, skin hyperpigmentation. Minimal myelosuppression.
Dactinomycin (actinomycin D)	Intercalates in DNA.	Wilms tumor, Ewing sarcoma, rhabdomyosarcoma. Used for childhood tumors.	Myelosuppression.
Doxorubicin, daunorubicin	Generate free radicals. Intercalate in DNA → breaks in DNA → ↓ replication.	Solid tumors, leukemias, lymphomas.	Cardiotoxicity (dilated cardiomyopathy), myelosuppression, alopecia. Dexrazoxane (iron chelating agent), used to prevent cardiotoxicity.

Alkylating agents

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Busulfan	Cross-links DNA.	CML. Also used to ablate patient's bone marrow before bone marrow transplantation.	Severe myelosuppression (in almost all cases), pulmonary fibrosis, hyperpigmentation.
Cyclophosphamide, ifosfamide	Cross-link DNA at guanine N-7. Require bioactivation by liver. A nitrogen mustard.	Solid tumors, leukemia, lymphomas.	Myelosuppression; hemorrhagic cystitis, prevented with mesna (thiol group of mesna binds toxic metabolites) or N-acetylcysteine.
Nitrosoureas	Require bioactivation. Cross blood-brain barrier → CNS. Cross-link DNA.	Brain tumors (including glioblastoma multiforme).	CNS toxicity (convulsions, dizziness, ataxia).

Microtubule inhibitors

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Paclitaxel, other taxols	Hyperstabilize polymerized microtubules in M phase so that mitotic spindle cannot break down (anaphase cannot occur).	Ovarian and breast carcinomas.	Myelosuppression, neuropathy, hypersensitivity.
Vincristine, vinblastine	Vinca alkaloids that bind β -tubulin and inhibit its polymerization into microtubules → prevent mitotic spindle formation (M-phase arrest).	Solid tumors, leukemias, Hodgkin (vinblastine) and non-Hodgkin (vincristine) lymphomas.	Vincristine: neurotoxicity (areflexia, peripheral neuritis), constipation (including paralytic ileus).

Cisplatin, carboplatin

MECHANISM	Cross-link DNA.
CLINICAL USE	Testicular, bladder, ovary, and lung carcinomas.
ADVERSE EFFECTS	Nephrotoxicity, peripheral neuropathy, ototoxicity. Prevent nephrotoxicity with amifostine (free radical scavenger) and chloride (saline) diuresis.

Etoposide, teniposide

MECHANISM	Etoposide inhibits topoisomerase II → ↑ DNA degradation.
CLINICAL USE	Solid tumors (particularly testicular and small cell lung cancer), leukemias, lymphomas.
ADVERSE EFFECTS	Myelosuppression, alopecia.

Irinotecan, topotecan

MECHANISM	Inhibit topoisomerase I and prevent DNA unwinding and replication.
CLINICAL USE	Colon cancer (irinotecan); ovarian and small cell lung cancers (topotecan).
ADVERSE EFFECTS	Severe myelosuppression, diarrhea.

Hydroxyurea

MECHANISM	Inhibits ribonucleotide reductase → ↓ DNA Synthesis (S-phase specific).
CLINICAL USE	Myeloproliferative disorders (eg, CML, polycythemia vera), sickle cell (↑ HbF), melanoma.
ADVERSE EFFECTS	Severe myelosuppression.

Bevacizumab

MECHANISM	Monoclonal antibody against VEGF. Inhibits angiogenesis (BeV acizumab inhibits B lood V essel formation).
CLINICAL USE	Solid tumors (colorectal cancer, renal cell carcinoma).
ADVERSE EFFECTS	Hemorrhage, blood clots, and impaired wound healing.

Erlotinib

MECHANISM	EGFR tyrosine kinase inhibitor.
CLINICAL USE	Non-small cell lung carcinoma.
ADVERSE EFFECTS	Rash.

Cetuximab

MECHANISM	Monoclonal antibody against EGFR.
CLINICAL USE	Stage IV colorectal cancer (wild-type <i>KRAS</i>), head and neck cancer.
ADVERSE EFFECTS	Rash, elevated LFTs, diarrhea.

Imatinib

MECHANISM	Tyrosine kinase inhibitor of <i>BCR-ABL</i> (Philadelphia chromosome fusion gene in CML) and <i>c-kit</i> (common in GI stromal tumors).
CLINICAL USE	CML, GI stromal tumors (GIST).
ADVERSE EFFECTS	Fluid retention.

Rituximab

MECHANISM	Monoclonal antibody against CD20, which is found on most B-cell neoplasms.
CLINICAL USE	Non-Hodgkin lymphoma, CLL, ITP, rheumatoid arthritis.
ADVERSE EFFECTS	↑ risk of progressive multifocal leukoencephalopathy.

Bortezomib, carfilzomib

MECHANISM	Proteasome inhibitors, induce arrest at G2-M phase and apoptosis.
CLINICAL USE	Multiple myeloma, mantle cell lymphoma.
ADVERSE EFFECTS	Peripheral neuropathy, herpes zoster reactivation.

Tamoxifen, raloxifene

MECHANISM	Selective estrogen receptor modulators (SERMs)—receptor antagonists in breast and agonists in bone. Block the binding of estrogen to ER ⊕ cells.
CLINICAL USE	Breast cancer treatment (tamoxifen only) and prevention. Raloxifene also useful to prevent osteoporosis.
ADVERSE EFFECTS	Tamoxifen—partial agonist in endometrium, which ↑ the risk of endometrial cancer; “hot flashes.” Raloxifene —no ↑ in endometrial carcinoma (so you can relax!), because it is an estrogen receptor antagonist in endometrial tissue. Both ↑ risk of thromboembolic events (eg, DVT, PE).

Trastuzumab (Herceptin)

MECHANISM	Monoclonal antibody against HER-2 (<i>c-erbB2</i>), a tyrosine kinase receptor. Helps kill cancer cells that overexpress HER-2, through inhibition of HER2-initiated cellular signaling and antibody-dependent cytotoxicity.
CLINICAL USE	HER-2 ⊕ breast cancer and gastric cancer (tras 2 zumab).
ADVERSE EFFECTS	Cardiotoxicity. “ Heart ceptin” damages the heart .

Vemurafenib

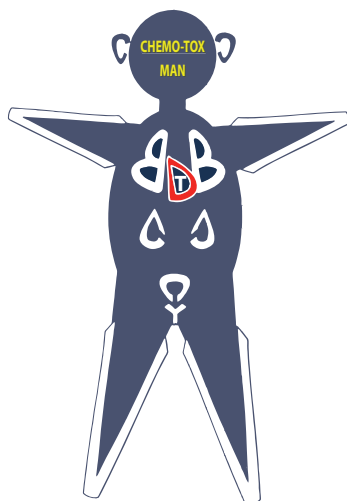
MECHANISM	Small molecule inhibitor of <i>BRAF</i> oncogene ⊕ melanoma. VEmuRAF-enib is for V600E -mutated BRAF inhibition.
CLINICAL USE	Metastatic melanoma.

Tumor lysis syndrome

Oncologic emergency triggered by massive tumor cell lysis, most often in lymphomas/leukemias. Release of K^+ → hyperkalemia, release of PO_4^{3-} → hyperphosphatemia, hypocalcemia due to Ca^{2+} sequestration by PO_4^{3-} . ↑ nucleic acid breakdown → hyperuricemia → acute kidney injury. Treatments include aggressive hydration, allopurinol, rasburicase.

Rasburicase

MECHANISM	Recombinant uricase that catalyzes metabolism of uric acid to allantoin.
CLINICAL USE	Prevention and treatment of tumor lysis syndrome.

**Common
chemotoxicities**

Cisplatin/**C**arboplatin → ototoxicity (and nephrotoxicity)

Vincristine → peripheral neuropathy

Bleomycin, **B**usulfan → pulmonary fibrosis

Doxorubicin → cardiotoxicity

Trastuzumab (Herceptin) → cardiotoxicity

Cisplatin/**C**arboplatin → nephrotoxicity (and ototoxicity)

CYclophosphamide → hemorrhagic cystitis